Note: to ensure compatibility with multiboard designs,
true 3v3 rail (referenced to ground) must be provided on
the subsequent board to provide ground reference to hall
effect sensors and mosfet drivers.

Ground rail also to be provided to subsequent board to
ensure common references for control signals

Alternatively, run 1 nucleo per board

3v3 and Ground output to next board

J1

Conn_02x02_Odd_Even

+3V3

GND

Spare connector to interact with following board

XT60_4

Shuffler Output

String output

PV1+ Sense

PV1+

XT60_1

PV Connector 1

PV1−

PV2+ Sense

PV2+

XT60_2

PV Connector 2

PV3+ Sense

PV3+

XT60_3

PV Connector 3

GND

Shuffle Converter 1

Input_T

Output

Input_B

Floating 3v3

PWM_A_H

PWM_A_L

PWM_B_H

PWM_B_L

I_Sense_Out_Aph

I_Sense_Out_Bph

S1_A_H

S1_A_L

S1_B_H

S1_B_L

S1_Aph_Sense

S1_Bph_Sense

File: Shuffle_Converter_Top.kicad_sch

Floating 3v3

Shuffle Converter 2

S2_A_H

S2_A_L

S2_B_H

S2_B_L

S2_Aph_Sense

S2_Bph_Sense

PWM_A_H

PWM_A_L

PWM_B_H

PWM_B_L

I_Sense_Out_Aph

I_Sense_Out_Bph

Input_T

Output

Input_B

File: Shuffle_Converter.kicad_sch

Voltage Sensing

Floating 3v3

MISO Bus

MOSI Bus

SCLKBus

Floating 3v3

CS1 Select

CS2 Select

CS3 Select

File: V_Sense.kicad_sch

Connections to be changed, confirm new pinout arrangment
with less gate drive signals but new SPI integration

U1A

NUCLEO-G474RE

S4_A_L

Vpv1

Vpv2

ILA_1

ILA_2

Vpv3

Vstring

ILA_4

ILA_3

U1B

NUCLEO-G474RE

S3_B_L

S4_B_H

ILB_1

ILB_2

S4_A_H

S3_A_L

S1_A_L

S1_A_H

S4_B_L

S1_B_H

ILB_3

ILB_4

S3_B_H

S3_A_H

S1_B_L

S2_A_H

Vpv5

Vpv4

Istring

S2_B_L

S2_B_H

S2_A_L

Sheet: /

File: Version 0 project.kicad_sch

Title:

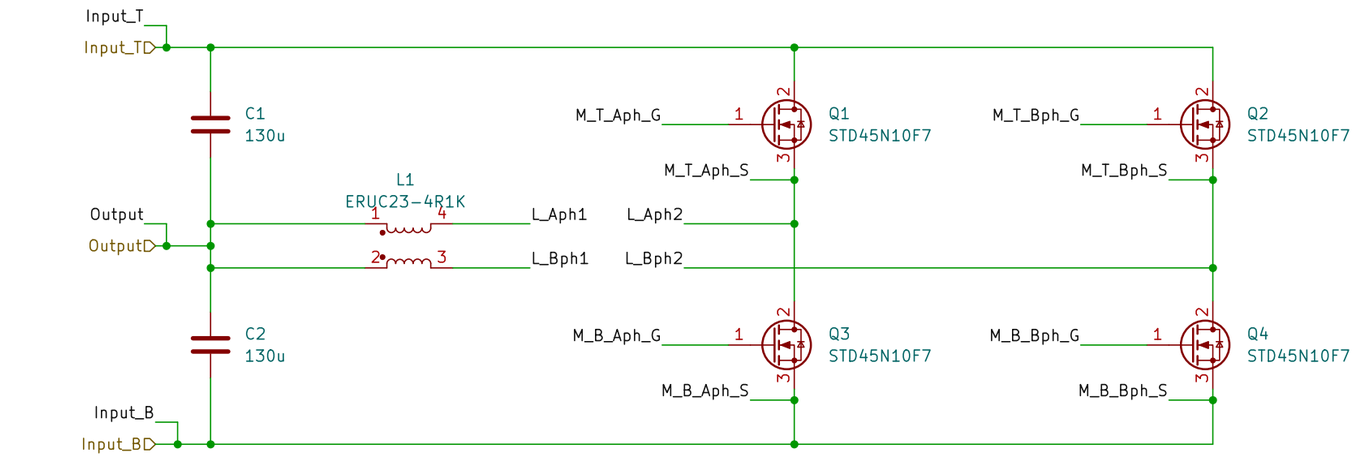
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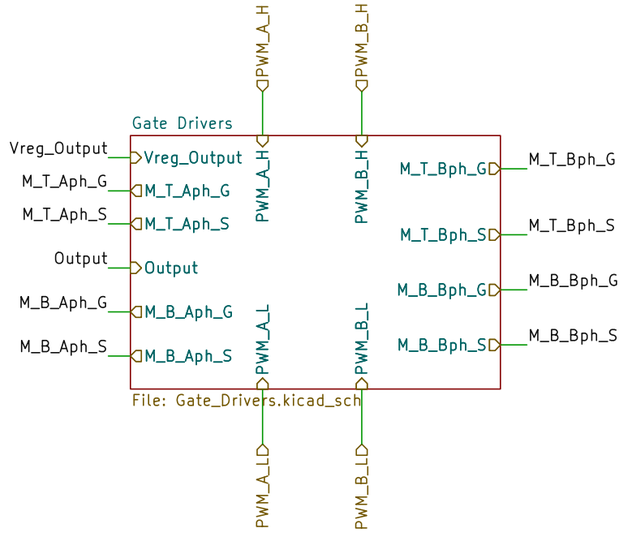
Date:

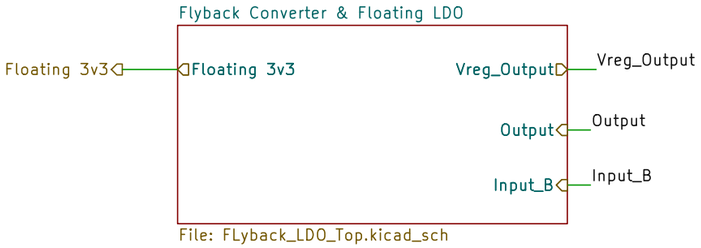
Rev:

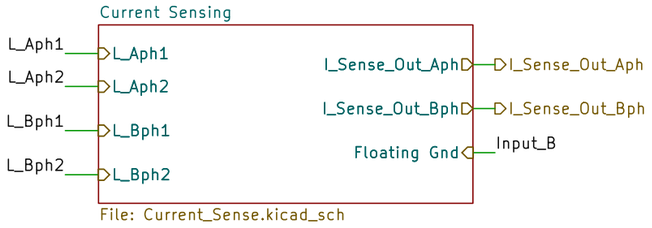
KiCad E.D.A. 10.0.5

Id: 1/10

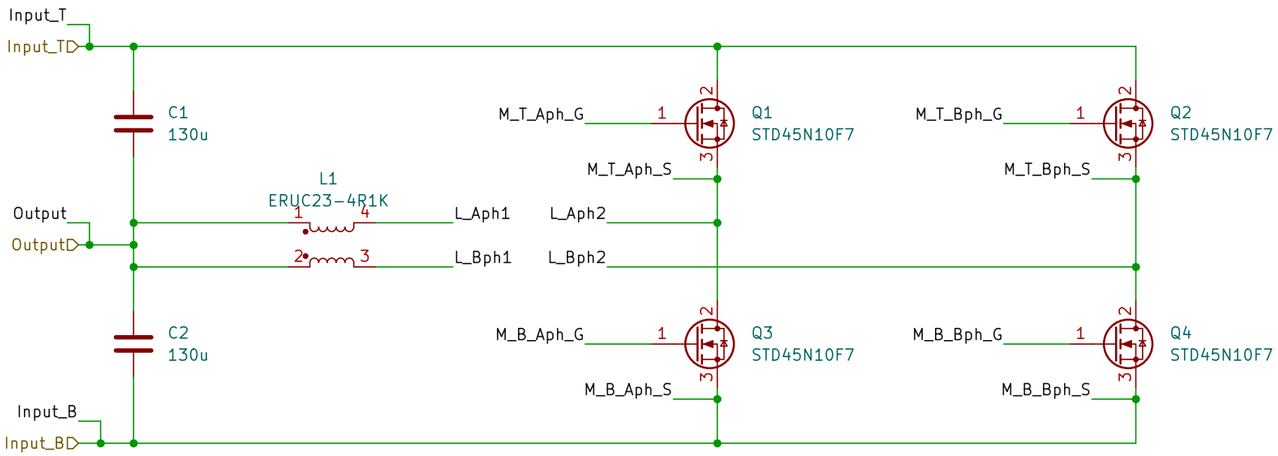


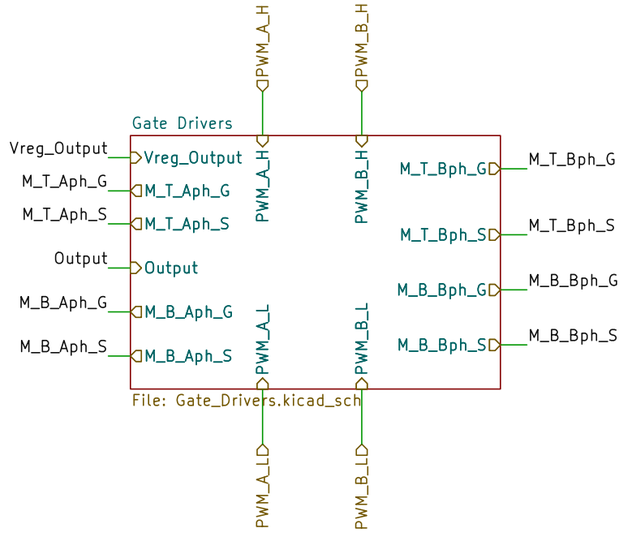


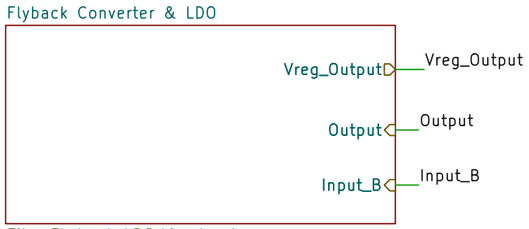


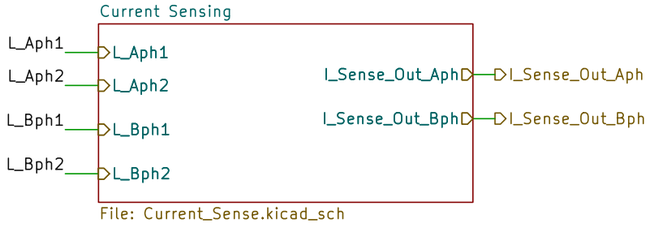


Sheet: /Shuffle Converter 1/		
File: Shuffle_Converter_Top.kicad_sch		
Title: Shuffler 1		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.5		Id: 2/10

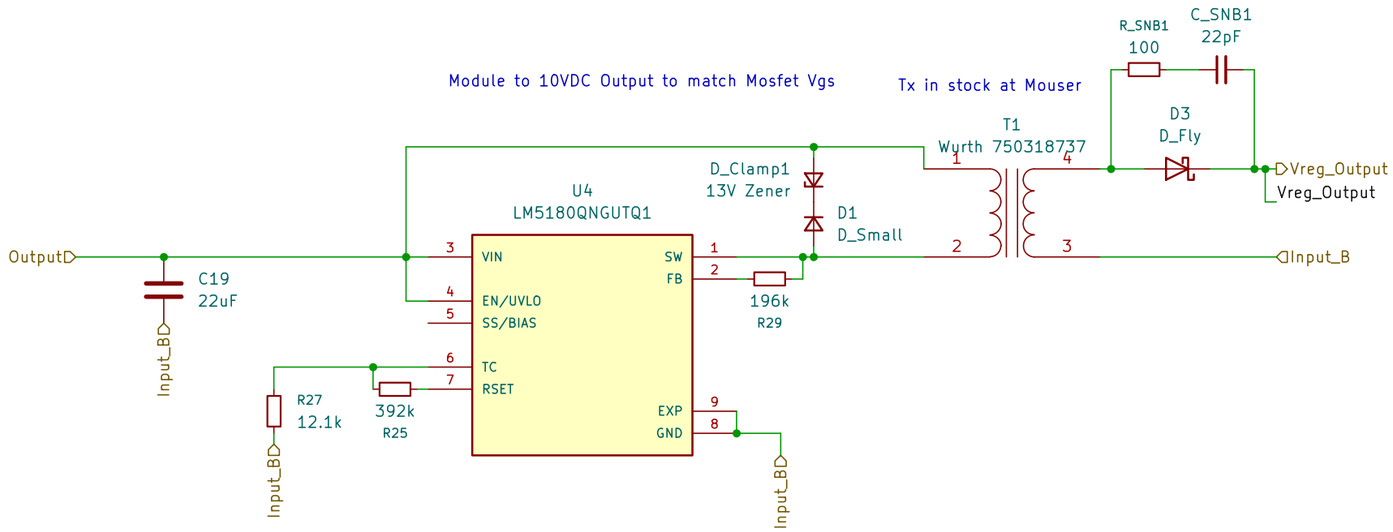


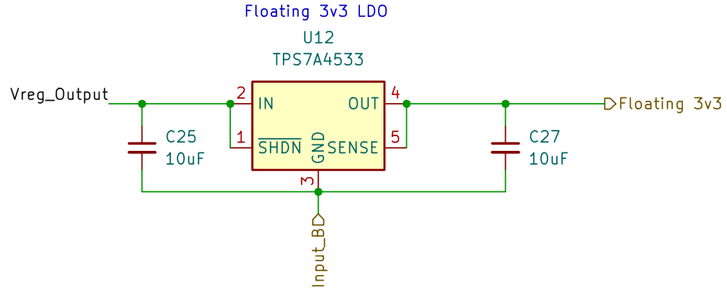






Sheet: /Shuffle Converter 2/		
File: Shuffle_Converter.kicad_sch		
Title: Shuffler 2		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.5		Id: 3/10

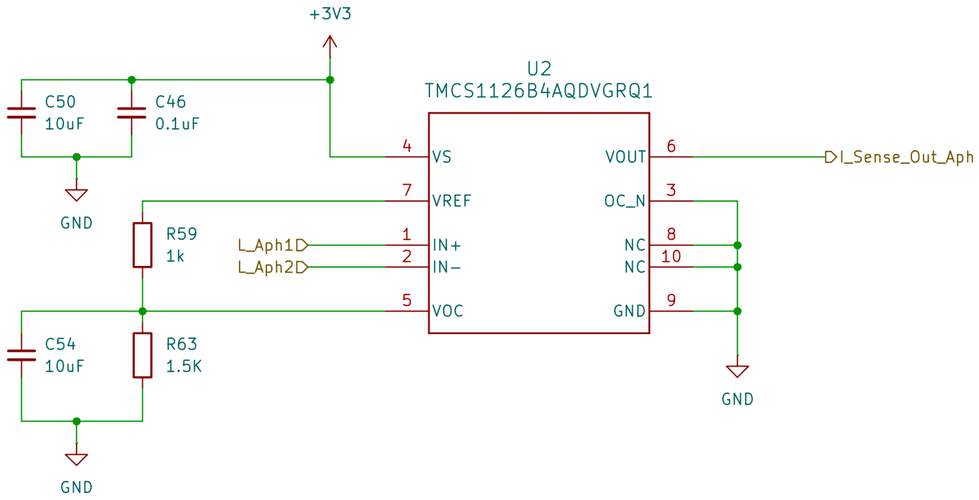


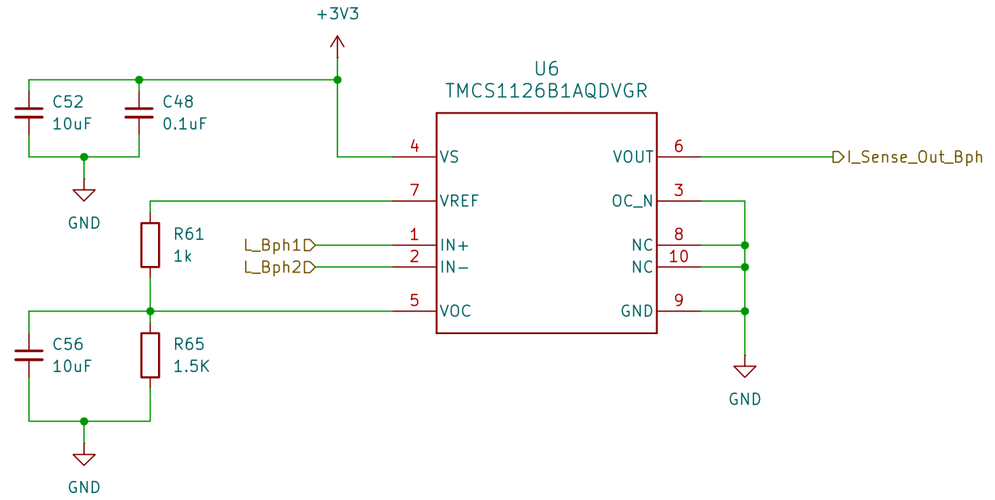


Sheet: /Shuffle Converter 1/Flyback Converter & Floating LDO/
File: FLyback_LDO_Top.kicad_sch

Title: Shuffler 1 Flyback and LDO

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.5		Id: 4/10

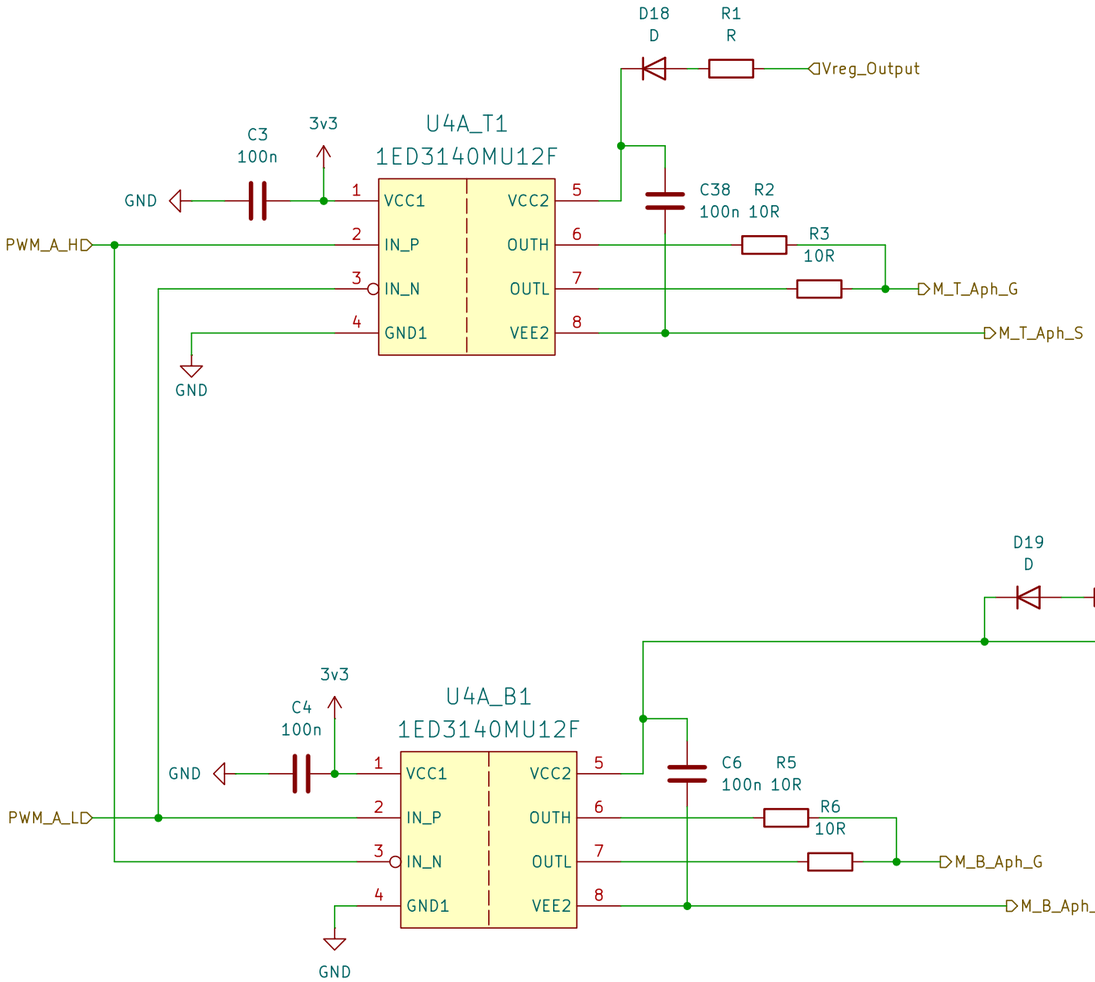


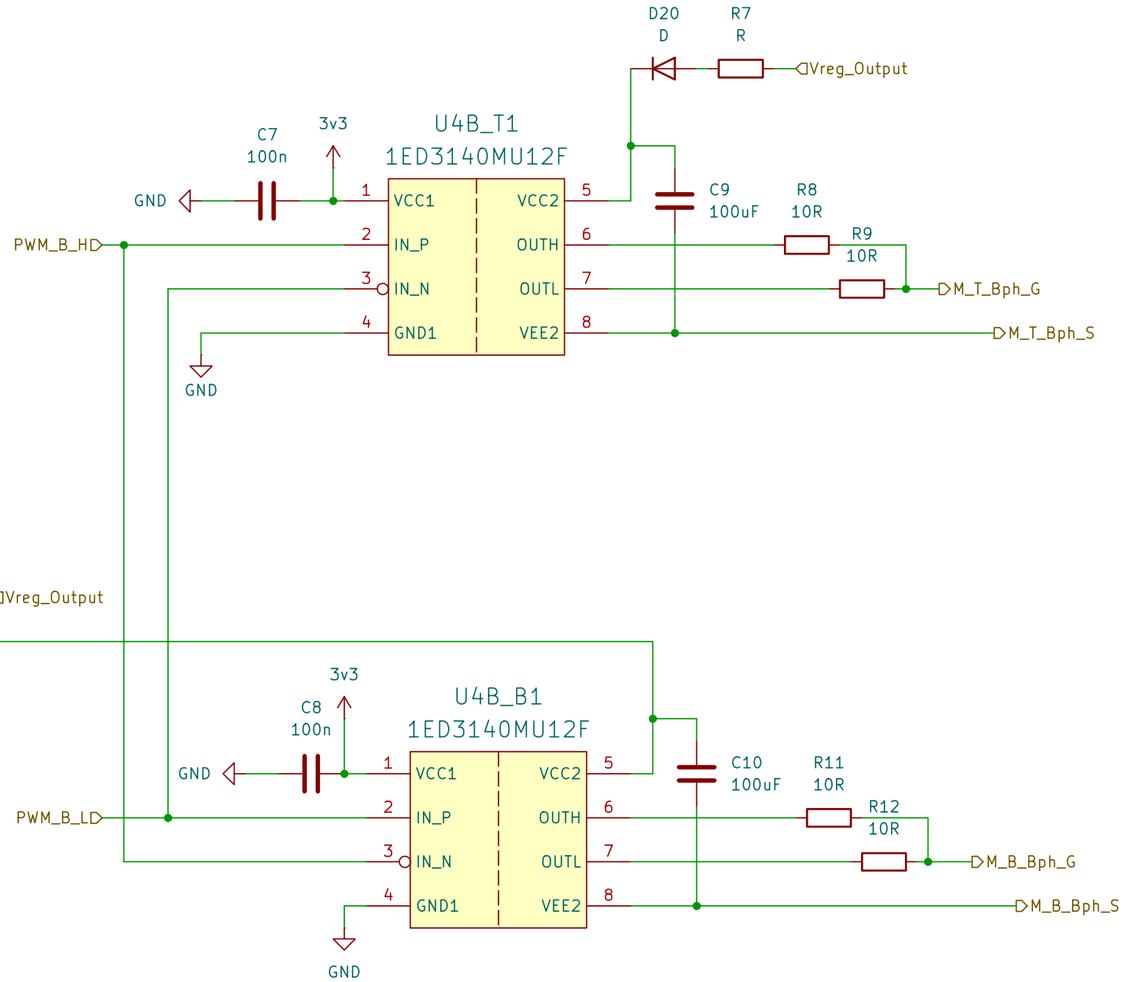


Sheet: /Shuffle Converter 1/Current Sensing/
File: Current_Sense.kicad_sch

Title: Inductor Current Sensing

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.5		Id: 5/10



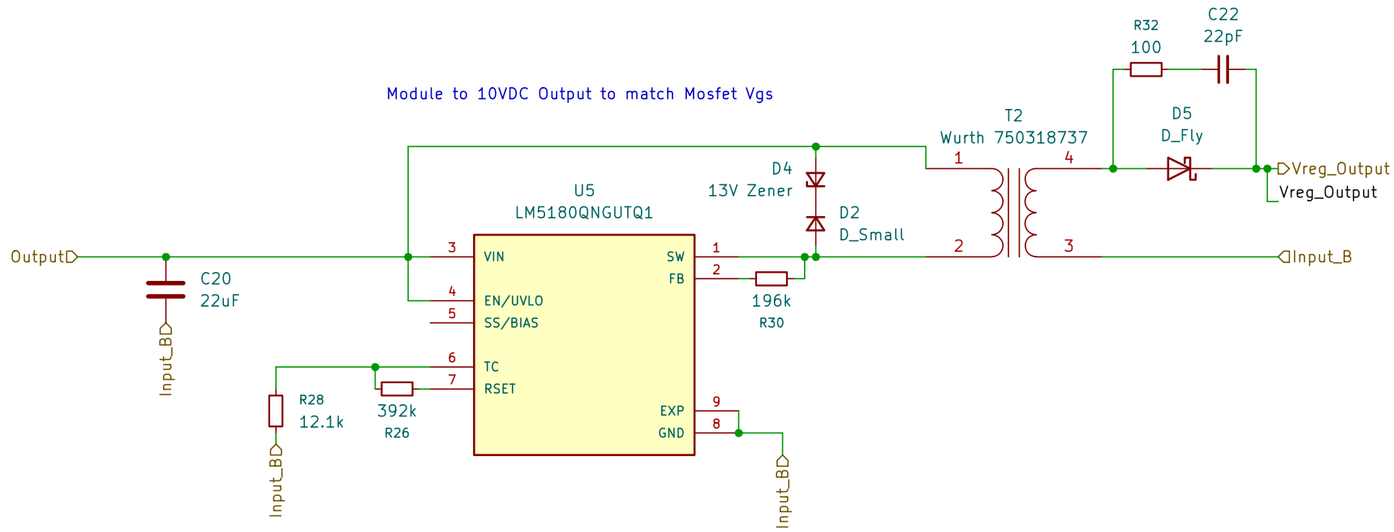


Sheet: /Shuffle Converter 1/Gate Drivers/
File: Gate_Drivers.kicad_sch

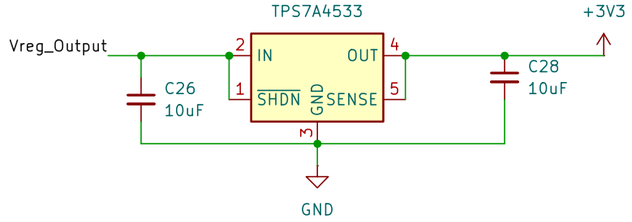
Title: Gate Drivers

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.5		Id: 6/10

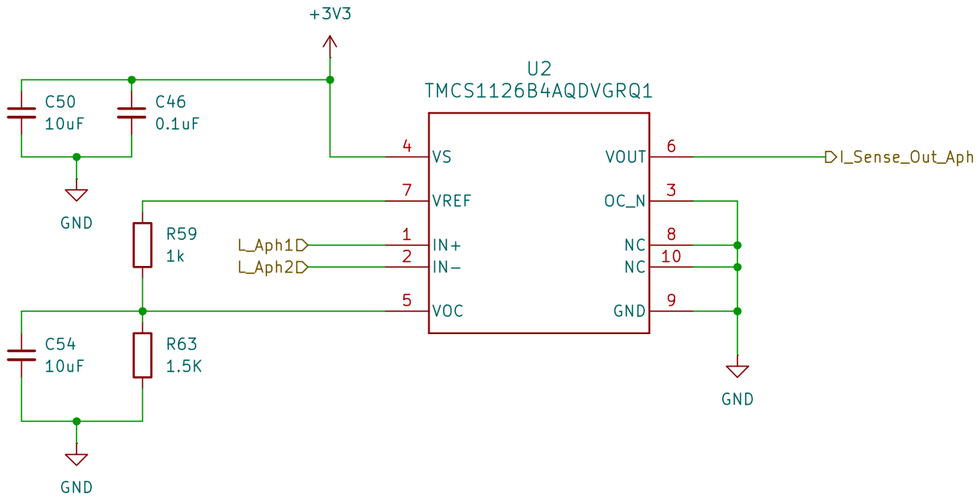
Module to 10VDC Output to match Mosfet Vgs

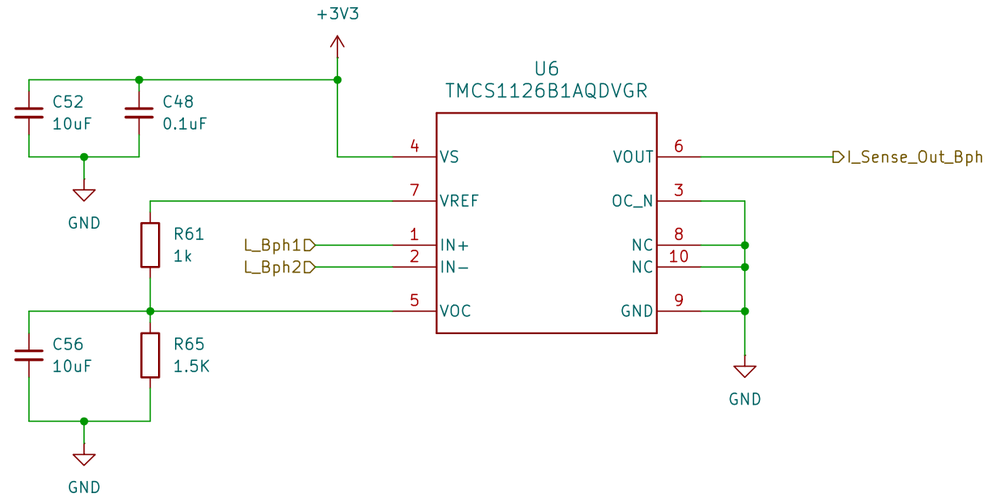


True 3v3 LDO



Sheet: /Shuffle Converter 2/Flyback Converter & LDO/		
File: FLyback_LDO.kicad_sch		
Title: Shuffler 2 Flyback and True 3v3 LDO		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.5		Id: 7/10

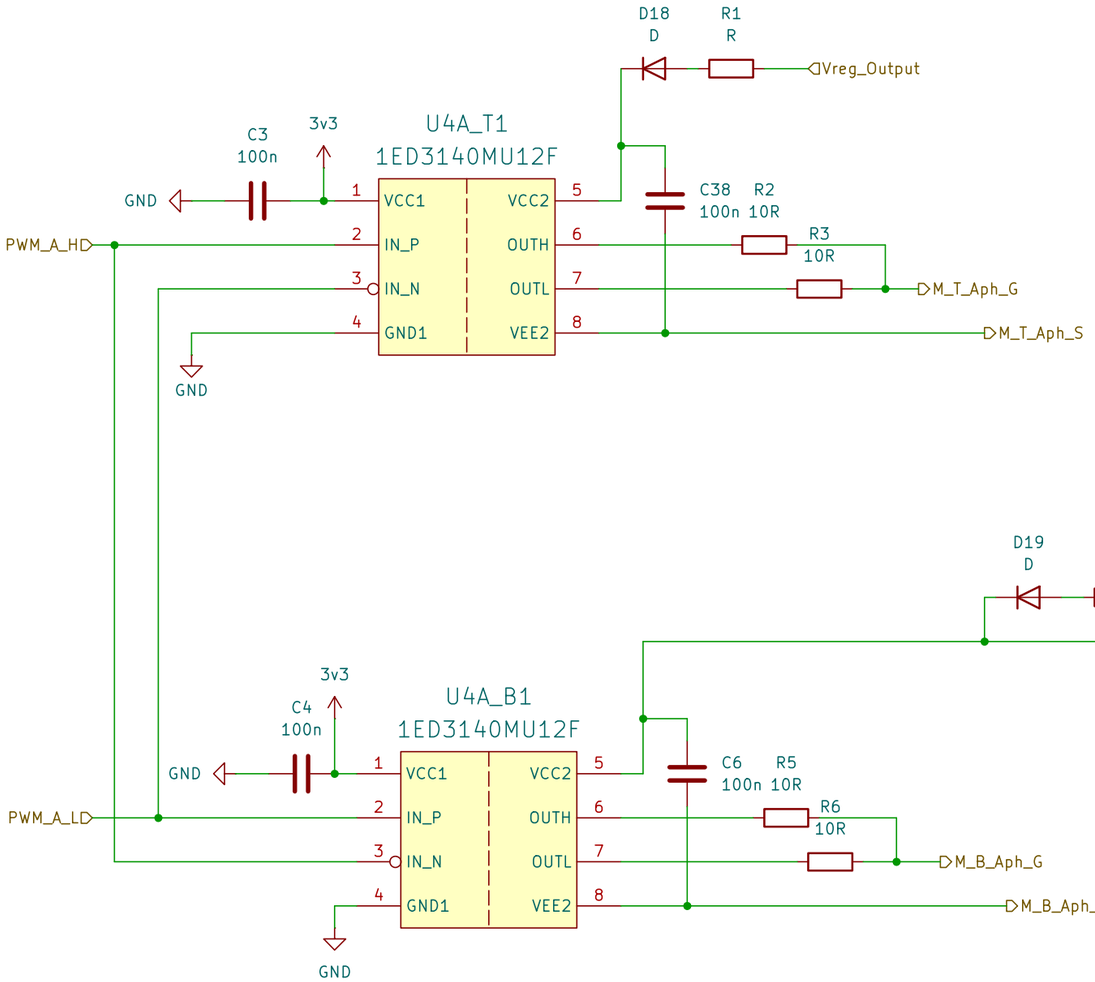


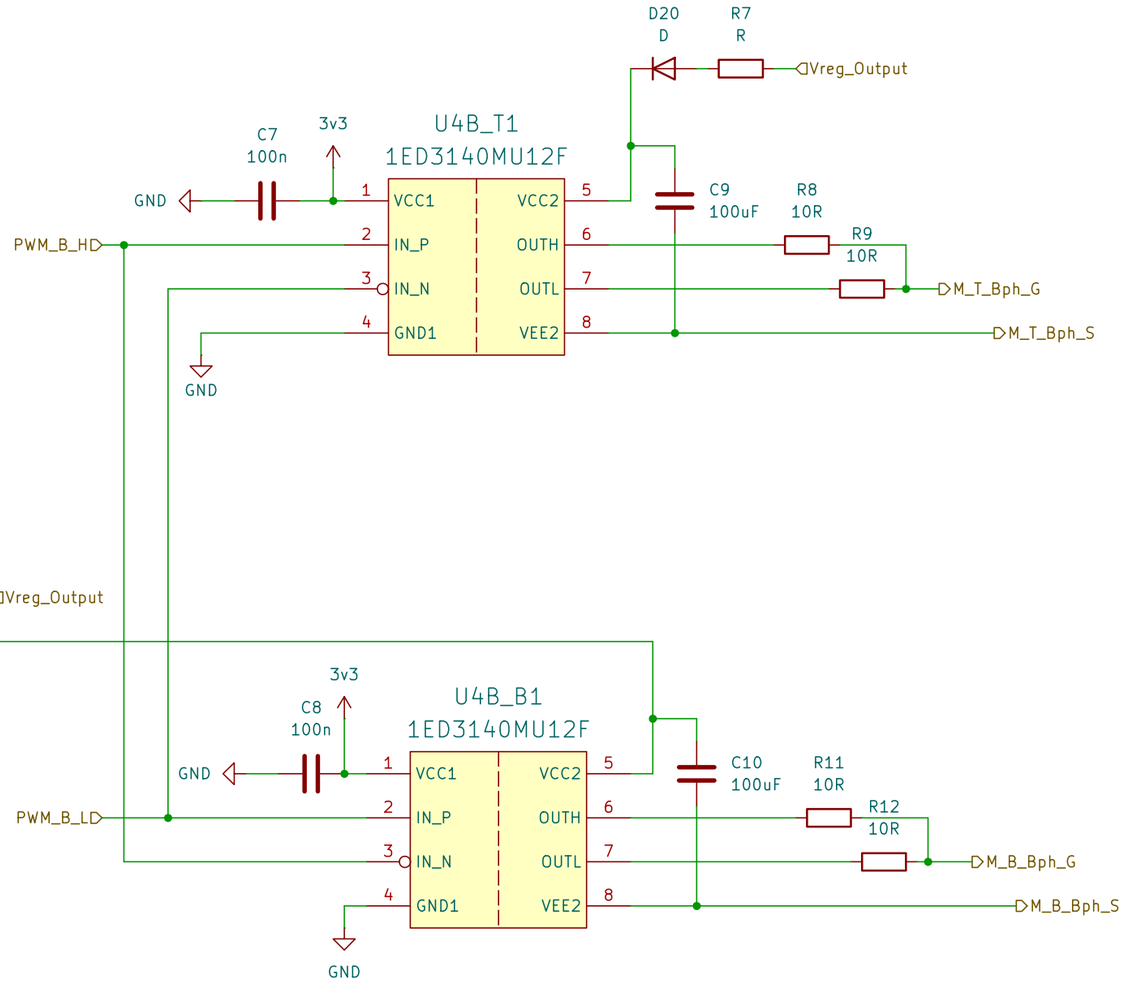


Sheet: /Shuffle Converter 2/Current Sensing/
File: Current_Sense.kicad_sch

Title: Inductor Current Sensing

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.5		Id: 8/10



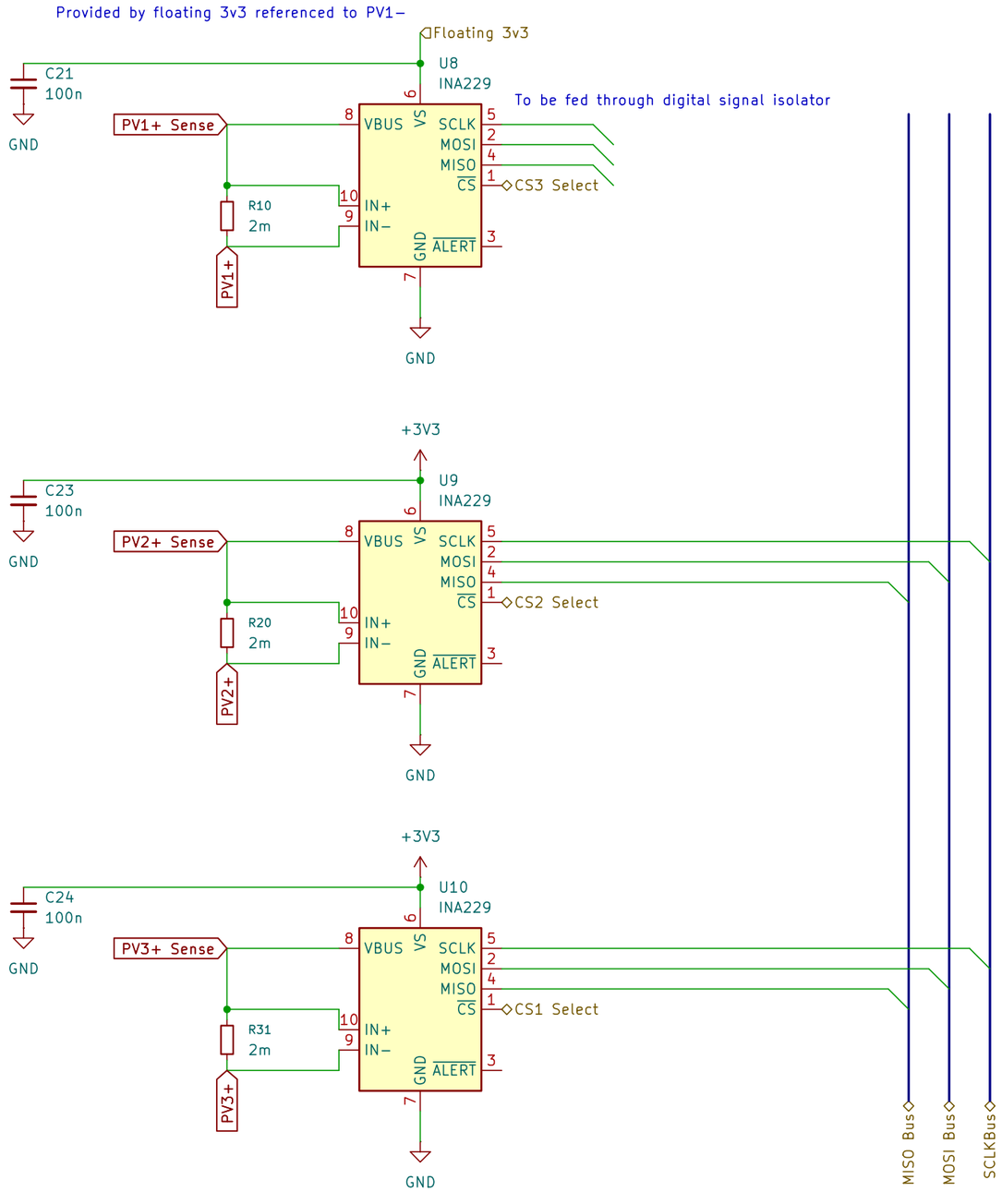


Sheet: /Shuffle Converter 2/Gate Drivers/
File: Gate_Drivers.kicad_sch

Title: Gate Drivers

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.5		Id: 9/10

Note: INA229s will also provide module current measurement



Sheet: /Voltage Sensing/	
File: V_Sense.kicad_sch	
Title: Module Voltage and Current Sensing	
Size: A4	Date:
KiCad E.D.A. 10.0.5	Id: 10/10